Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Automate Business Processes with Agentic AI Workflows

Course Code: C209

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.5

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	8 August 2026	Derived from the parent course courseware (v2.0): same topics, labs and house design.	Dr. Alfred Ang
2.0	19 September 2026	Non-WSQ edition: assessment and funding content removed; both days retimed to 9:30am – 5:30pm (7.5 instructional hours/day, 15 hours total) with a 30-minute lunch break; slide references remapped.	Dr. Alfred Ang
2.5	17 October 2026	Updated from the parent course courseware v2.5: Topic 1 opens with the modern view of agentic automation (a brief history of AI, agentic AI vs AI agents, workflow first vs agent first, AI-assisted workflow creation with n8n Assistant, AI security); every lab now follows the concepts it needs; a Telegram HR chatbot demo before Lab 3; guardrails before Lab 4; real n8n screenshots explained for Labs 2–5. Assessment and funding content removed; both days 9:30am – 5:30pm (7.5 instructional hours/day) with a 30-minute lunch break; slide references remapped.	Dr. Alfred Ang

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Course Information

Course Title	Automate Business Processes with Agentic AI Workflows
Course Code	C209
Course Type	Non-WSQ commercial short course
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	Day 1: 9:30 am – 5:30 pm. Day 2: 9:30 am – 5:30 pm. A 30-minute lunch break each day is excluded from instructional time.
Mode	Instructor-led, hands-on workflow-automation labs built in n8n
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what n8n and AI agents are, and build automated workflows using triggers and actions.
- LO2: Implement AI agent applications that are triggered by webhooks and integrated with business systems.
- LO3: Design agentic chatbot workflows that operate safely within business and regulatory constraints.
- LO4: Apply human-in-the-loop approval so that sensitive actions are authorised by a person.
- LO5: Deploy Retrieval-Augmented Generation (RAG) to ground an agent in an organisation's own documents.

Course Schedule

Day 1 · 9:30am – 5:30pm — n8n, AI Agents and Agentic Chatbots

Time	Duration	Topic / Activity (with slide references)
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, course portal and workflows, course outline and learning outcomes. Slides 2–11
10:00–11:00	60 min	**Topic 1 — Overview of n8n and AI Agents: why agentic AI, and why workflow first — a brief history of AI 2023–**

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		2026, generative AI vs agentic AI, agentic AI vs AI agents, workflow first vs agent first, AI-assisted workflow creation, platform tiers (no-code, AI-assisted low-code, AI-generated full-code), business value, use cases, the five principles of AI security, what n8n is, signing in to n8n, and a demo that builds a social media campaign workflow with n8n Assistant. Slides 12–28
11:00–11:30	30 min	**Topic 1 (continued) — n8n fundamentals: trigger nodes, data-processing nodes, storage and app actions, JSON and expressions, test vs production executions, and credentials. Slides 29–38**
11:30–12:15	45 min	Hands-on Lab 1 — n8n Trigger and Action (Event Flyer with QR Code): build the form-to-Gmail workflow, publish it, copy the form's Production URL and create a QR-code event flyer; present and scan-test the flyers. Slides 39–47
12:15–12:45	30 min	Lunch break
12:45–14:30	105 min	**Topic 1 (continued) — The building blocks of an AI agent, before Lab 2: the AI agent and its loop, the model (popular LLM providers, model parameters and temperature, claiming the free AI credits), tools and MCP, instructions with worked examples, memory, and structured output. Slides 49–61**
14:30–15:15	45 min	**Topic 1 (continued) — Webhooks: the inbound URL, webhook vs HTTP Request, GET vs POST, Test vs Production URL and webhook configuration. Slides 62–69**
15:15–17:15	120 min	Hands-on Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding): how the webhook, agent, model, tools, instructions, memory and structured output fit together, then build and test the onboarding agent. Slides 70–74

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17:15–17:30	15 min	Day 1 recap and Q&A

Instructional time: 450 minutes (7.5 hours). The day runs 9:30 am – 5:30 pm: the 30-minute lunch break is excluded from instructional time.

Day 2 · 9:30am – 5:30pm — Human in the Loop and RAG

Time	Duration	Topic / Activity (with slide references)
9:30–9:45	15 min	Day 1 recap
9:45–10:30	45 min	**Topic 2 — Agentic Chatbot and Human in the Loop, before Lab 3: rule-based vs agentic workflows, designing a safe agentic chatbot, and a chatbot demo on Telegram — learners create a bot with BotFather /newbot, build an HR policy chatbot with n8n Assistant from one prompt, then connect, publish and chat with it. Slides 75–84**
10:30–11:00	30 min	Hands-on Lab 3 — Chatbot Webhook (Investment Advisor): import the workflow, read the contact gate, the non-advisory rule and the FAQ, and deploy the chatbot behind the website. Slides 85–88
11:00–11:30	30 min	Hands-on Lab 3 — Chatbot Webhook (continued): test the contact gate and the four compliance probes, and debrief.
11:30–12:15	45 min	**Topic 2 (continued) — Human in the loop and guardrails, before Lab 4: human in, on and out of the loop; how an approval suspends a running workflow; when a human belongs in the workflow; and guardrails. Slides 89–93**
12:15–12:45	30 min	Lunch break
12:45–14:15	90 min	Hands-on Lab 4 — Human in the Loop (Client Rapport Assistant): the approve and decline paths. Slides 94–98
14:15–15:30	75 min	**Topic 3 — Retrieval-Augmented Generation: why retrieval rather than a bigger prompt, the RAG pipeline, tokenization and embeddings, chunking, vector stores and grounding. Slides 99–107**
15:30–17:00	90 min	Hands-on Lab 5 — RAG Customer Care Chatbot (Cook & Bake Academy): ingestion

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		and the grounded chatbot. Slides 108–114
17:00–17:30	30 min	Course recap, taking it back to work, Q&A. Slides 115–117

Instructional time: 450 minutes (7.5 hours). The day runs 9:30 am – 5:30 pm: the 30-minute lunch break is excluded from instructional time.

Total instructional time across the course: 15 hours (2 days × 7.5 hours).

Lab Reference (aligned to topics and learning outcomes)

Topic	Outcomes	Labs	Deck slides
Topic 01: Overview of n8n and AI Agents	LO1 · LO2	Lab 1, Lab 2	Slides 12–74
Topic 02: Agentic Chatbot and Human in the Loop	LO3 · LO4	Lab 3, Lab 4	Slides 75–98
Topic 03: Retrieval-Augmented Generation (RAG)	LO5	Lab 5	Slides 99–114

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